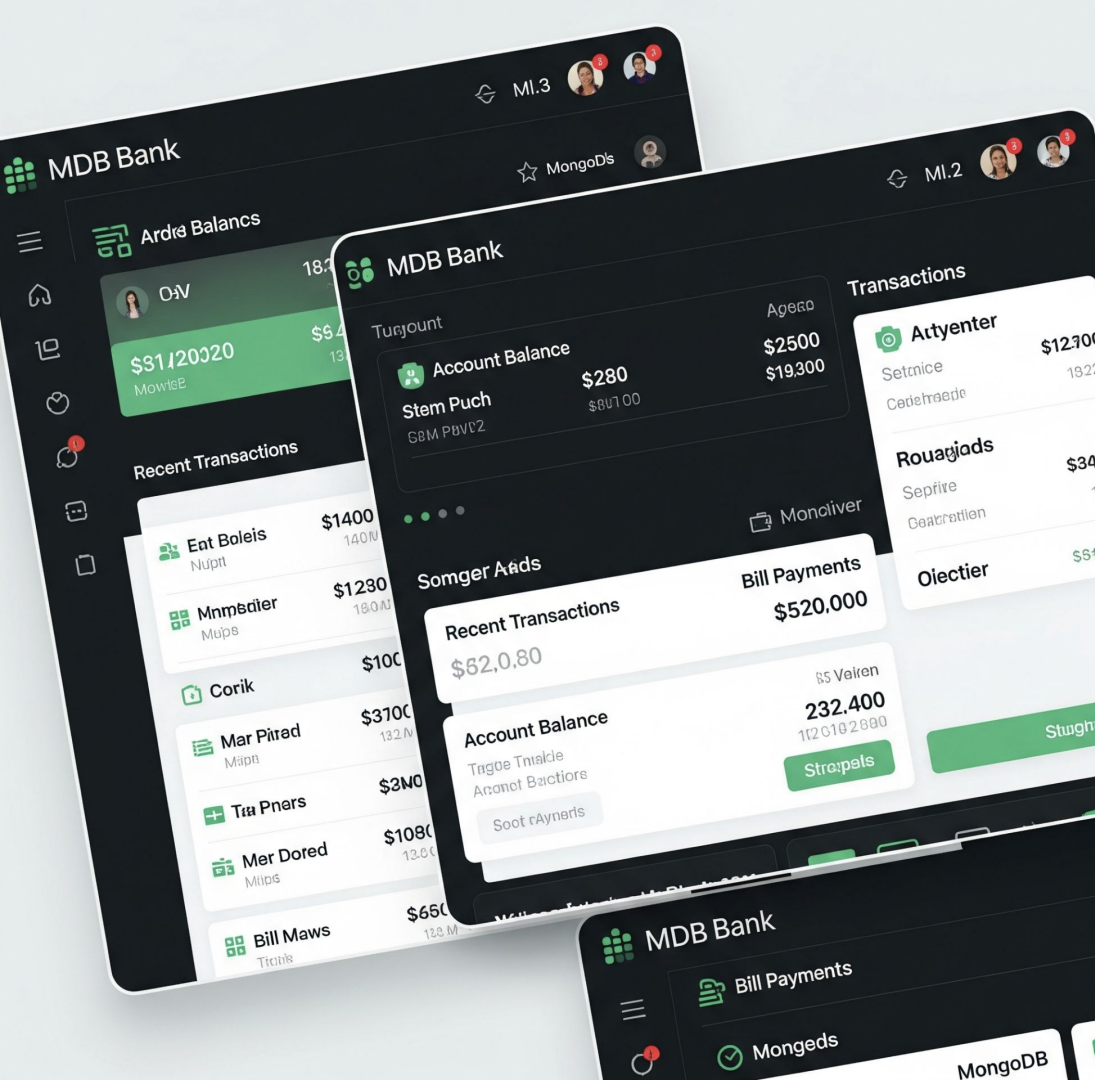




Scenario: MDB Bank App

Scenario: MDB Bank App



Customer
Overview

MDBBank

John Doe

Bank Accounts

123456789\$5,320.00

Checking

987654321\$10,245.50

Savings

View latest
transactions

\$5,320.00

Checking

1234567890

Latest Transactions

Apr 22Deposit\$1,200.00

Apr 20Grocery Store-\$55.40

Apr 19Payment-400.00

Apr 17Electric Bill-125.75

View Statements

View past statements up to 3 years

View past
statements

March 2024 Statement

Date	Description	Change	Balance
Mar 1	Balance B/F		\$2,800.00
Mar 5	ATM Withdraw	-\$100.00	\$2,700.00
Mar 10	SALARY	+\$5,800.00	\$8,500.00
Mar 15	BILL	-\$100.00	\$8,400.00
Mar 22	Txf Jane	-\$3500.00	\$4,900.00

View/Manage
Payees

MDBBank

Payees

Add

Alice BrownEditDelete

Carla GreenEditDelete

John SmithEditDelete

Robert WhiteEditDelete

MDBBank
Manage Recurring Transfers

Recurring Transfers

Add New Transfer

Frequency: Monthly

From: Account No. 1234

Last Transfer: Feb 15, 2024

To: John Doe

EditDelete

Frequency: Monthly

From: Account No. 5678

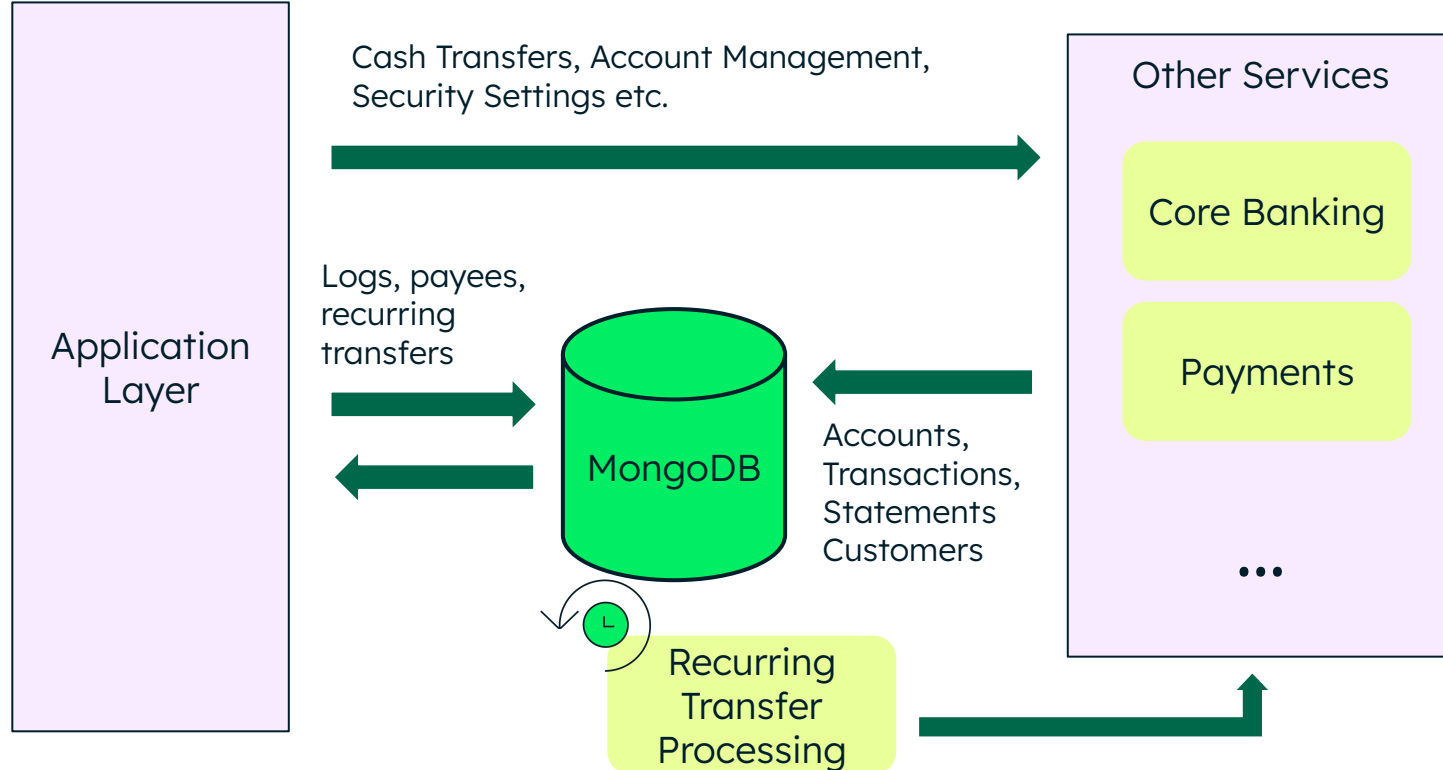
Last Transfer: Feb 15, 2024

To: John Doe

EditDelete

View/Manage
Recurring Transfers

High Level Architecture



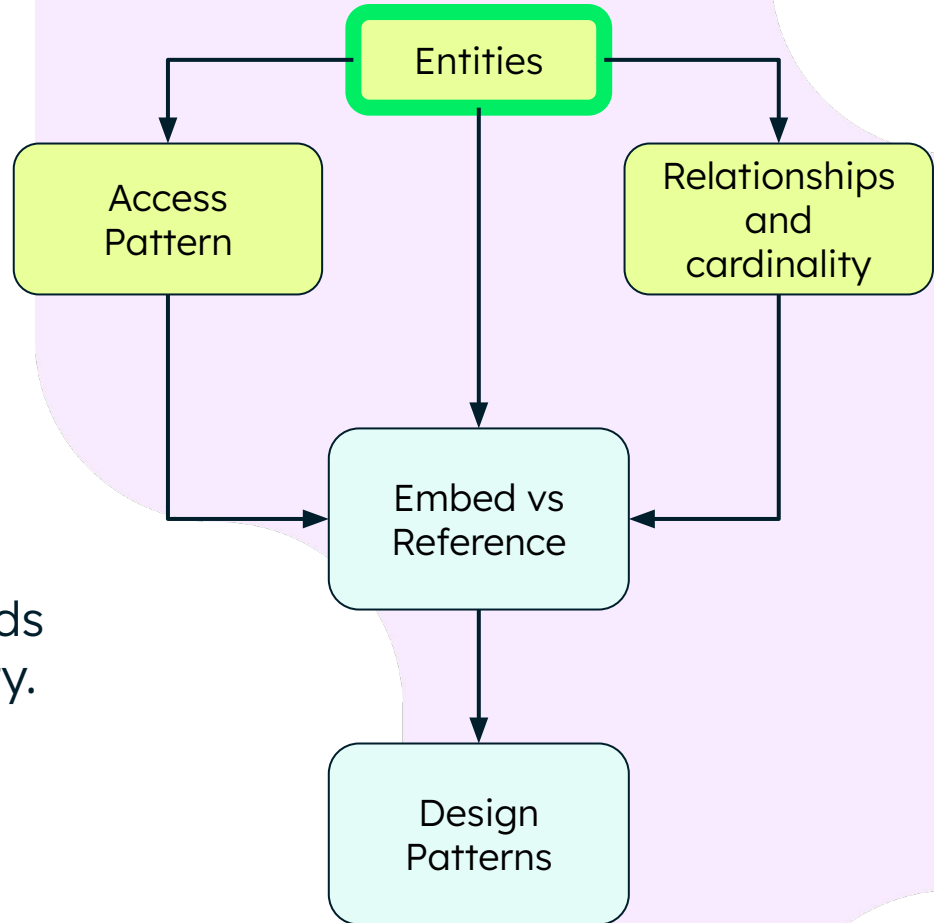


Exercise 1:

For each of the features, the entities has been identified for you.

Based on the screenshot and requirements, determine the fields required to be stored in the entity.

Exclude reference ids at the moment





Customer Overview

- Displays customer's first name and last name
- A customer can have multiple accounts

MDBBank

John Doe

Bank Accounts

123456789 \$5.320.00
Checking

987654321 \$10.245.50
Savings

Customer
ID: National ID Number
...

Account
ID: Bank account number
...



\$5,320.00

Checking

123-456-7891

Transfer limit \$5000.00

Latest Transactions

Apr 22 Deposit \$ 1,200.00

Apr 20 Grocery Store -\$55.40

Apr 19 Payment -400.00

Apr 17 Electric Bill -125.75

View Statements

View past statements up to 3 years

View account latest transactions

- Shows account transactions
- Latest transactions are ones that has not been saved in a statement

Account
ID: Bank account number
...

Transaction
ID: TransactionID
...



View statement

- View transactions from past statements of up to 3 years

March 2024 Statement

Date	Description	Change	Balance
Mar 1	Balance B/F		\$2,800.00
Mar 5	ATM Withdraw	-\$100,00	\$2,700.00
Mar 10	SALARY	+\$5,800.00	\$8,500.00
Mar 15	BILL	-\$100,00	\$8,400.00
Mar 22	Txf Jane	-\$3500,00	\$4,900.00

Statement

ID: StatementID

start_date

end_date

...

Line Item

ID: StatementID + Line No.

...

MDBBank

Payees

Add

Alice Brown

Edit

Delete

Carla Green

Edit

Delete

Add Payee

Payee Name

SWIFT code

Account No.

Save Payee

Payee Management



- CRUD of Payees

Payee
ID: PayeeID
...

MDBBank

Manage Recurring Transfers

Add New Transfer

Recurring Transfers

Frequency: Monthly From: Account No. 1234
Last Transfer: Feb 15, 2024 To: John Doe

Edit

Delete

Frequency: Monthly From: Account No. 5678
Last Transfer: Feb 15, 2024 To: John Doe

MDBBank

Add Recurring Transfer

Source Account Select Source Account

Payee Select Payee

Amount Enter Amount

Frequency Select Frequency

Start Date Select Start Date

End Date Select Start Date

Optional

Save Transfer

Cancel

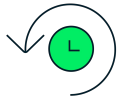
Recurring Transfers Management



- CRUD of recurring transfers
- Each recurring transfer allows user to set one source account and one payee account
- Frequency can be DAILY, WEEKLY, or MONTHLY

Recurring Transfer

ID: RecurringID

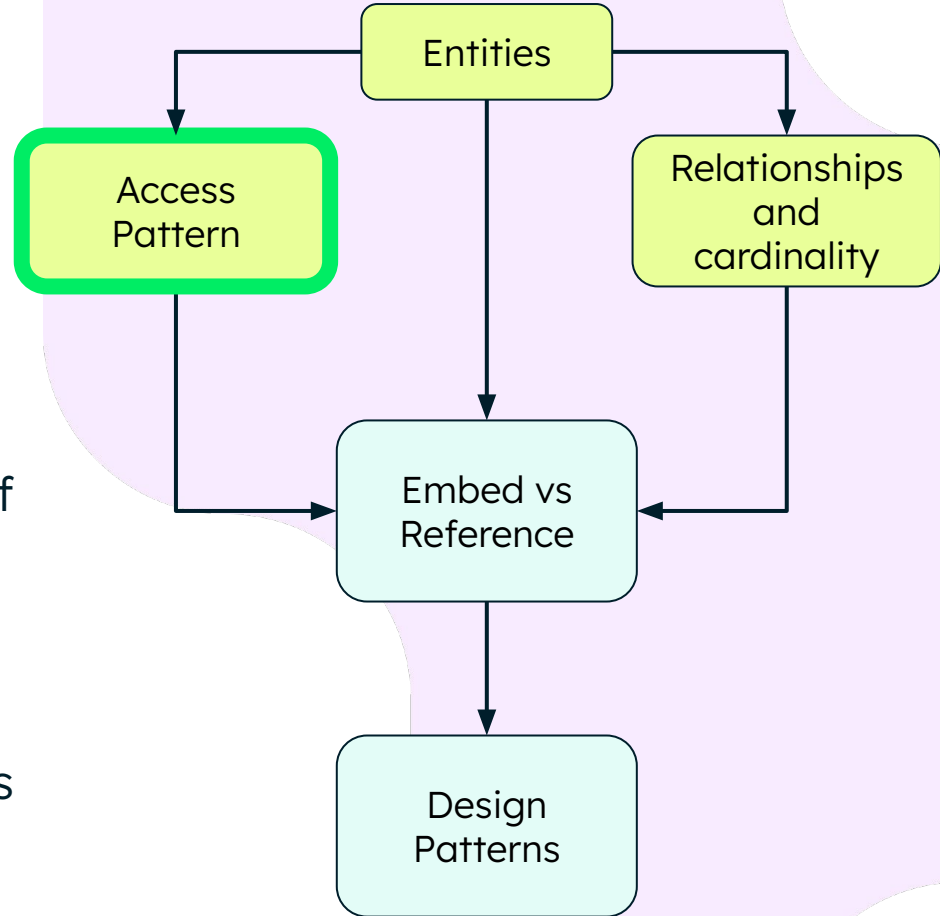


10 min

Exercise 2:

Based on the usage statistics given, calculate the frequency of operations per day for each entity.

Count a single operation that operates across multiple records as one operation.





Customer Overview

- 1 million active customers
- Every app access begins with this page
- An active customer accesses the app once every 2 days on average

Workloads

Operation	Ops per day
GET Customer	reads
GET Accounts	reads

MDBBank

John Doe

Bank Accounts

123456789 \$5.320.00

Checking

987654321 \$10.245.50

Savings



\$5,320.00

Checking

123-456-7891

Transfer limit \$5000.00

Latest Transactions

Apr 22 Deposit \$ 1,200.00

Apr 20 Grocery Store -\$55.40

Apr 19 Payment -400.00

Apr 17 Electric Bill -125.75

View Statements

View past statements up to 3 years

Read Account Details

- An active customer accesses each account once every 10 days
- An active customer holds 2 accounts on average
- “Latest Transactions” are ones that are not yet included in a statement

Workloads

Operation	Ops per day
GET Account	reads
GET Transaction	reads



\$5,320.00

Checking

123-456-7891

Transfer limit \$5000.00

Latest Transactions

Apr 22 Deposit \$ 1,200.00

Apr 20 Grocery Store -\$55.40

Apr 19 Payment -400.00

Apr 17 Electric Bill -125.75

View Statements

View past statements up to 3 years

Process Transaction

- Each account makes 1 transaction per day on average
- For every transaction, the account balance needs to be updated

Workloads

Operation	Ops per day
UPDATE Account	writes
CREATE Transaction	writes



\$5,320.00

Checking

123-456-7891

Transfer limit \$5000.00

Latest Transactions

Apr 22 Deposit \$ 1,200.00

Apr 20 Grocery Store -\$55.40

Apr 19 Payment -400.00

Apr 17 Electric Bill

March 2024 Statement

Date	Description	Change	Balance
Mar 1	Balance B/F		\$2,800.00
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Mar 15	BILL	-\$100.00	\$8,400.00
Mar 22	Txf Jane	-\$3500.00	\$4,900.00

View

View past st

Statements

- A new statement per account is written by an external service every month, and corresponding transactions are deleted
- Users access one statement from each account once a month
- Statements more than 3 years old are deleted

Workloads

Operations	Ops
GET Statement	reads per day
CREATE DELETE Statement	writes per month, deletes per month
DELETE Transactions by Account	bulk deletes per month

MDBBank

Payees

Add

Alice Brown

Edit

Delete

Carla Green

Edit

Delete

Add Payee

Payee Name

SWIFT code

Account No.

Save Payee

Payee Management



- Users access this page once a quarter
- User adds, edits or deletes a payee about twice a year

Operations	Ops per day
GET Payees	1 million / 90 = 11k
ADD, EDIT, DELETE Payees	1 million / 180 = 5.5k

*These translates to less than 1 per second

MDBBank

Manage Recurring Transfers

Add New Transfer

Recurring Transfers

Frequency: Monthly

From: Account No. 1234

Last Transfer: Feb 15, 2024

To: John Doe

Edit

Delete

Frequency: M

Last Transfer

MDBBank

Add Recurring Transfer

Source Account

Select Source Account

Payee

Select Payee

Amount

Enter Amount

Frequency

Select Frequency

Start Date

Select Start Date



End Date

Select Start Date



Optional

Save Transfer

Cancel

Recurring Transfers Management



- For each recurring transfer, the customer chooses one “source” account to pay one payee
- Users access this page twice a year
- User adds, edits or deletes a recurring transfer about once a year

Operations	Ops per day
GET Recurring GET Payee GET Account	1 million / 180 = 5.5k
ADD, EDIT, DELETE Recurring	1 million / 365 = 2.7k

Add New Transfer

Recurring Transfers

Frequency: Monthly
Last Transfer: Feb 15, 2024

From: Account No. 1234
To: John Doe

Edit

Delete

Frequency: M
Last Transfer:

MDBBank

Add Recurring Transfer

Source Account
Select Source Account

Payee
Select Payee

Amount
Enter Amount

Frequency
Select Frequency

Start Date
Select Start Date

End Date
Select Start Date

Optional

Save Transfer

Cancel

Recurring Transfers Processing



- A job reads recurring transfers which are due on a daily basis for processing
- Another job updates the statuses and dates for the next transfer
- An active customer has an average of 3 transfers due per month

Operations	Ops per day
GET Recurring	1 batch of 100k records
ADD, EDIT, DELETE Recurring	1 batch of 100k records